

The Mathematics of Well-Being, Politics, and Misalignment

A Mathematical Framework for Collective Welfare,
Governmental Optimization, and Coordination Failure

ETHAN JAHAN

April 2026

Abstract

We develop a compact, purely set-theoretic mathematical framework for reasoning about human well-being and societal structures. The foundational sections (Sections 1–4) provide a self-contained axiomatic re-derivation of central results from classical social welfare theory in the tradition of Bentham, Harsanyi, and Atkinson [16]: well-being is formalized as a real-valued function over individuals and time, the empirically established logarithmic relationship between wealth and well-being is adopted, and the suboptimality of unequal wealth distributions under concave welfare is proved via the arithmetic-geometric mean inequality. Building on these foundations, we derive a new intertemporal decomposition of normalized collective well-being (Lemma 4.10) that integrates per-capita resources, distributional inequality, and demographic weight into a single formula; this decomposition, which to our knowledge has not previously appeared in this unified form, provides a principled basis from which the core objectives of governmental policy—environmental stewardship, optimal redistribution, and the protection of human life—follow as direct logical consequences. We introduce a formal definition of *Moloch*—the coordination-failure gap—as a real-valued measure of governmental misalignment, and analyze the structural forces, including the existence of uncoordinated sovereign nations, that sustain it. Finally, we treat AI, within the human welfare objective, as a causal operator on the human system, arguing that AI misalignment is a component of the Moloch phenomenon. In particular, we identify three major AI-related threats to human well-being—AI-driven inequality, AI-amplified interstate violence, and self-preserving superintelligence—and argue that the third is structurally special because a single loss-of-control event would be irreversible.

Contents

1	The Individual and the Well-Being Function	3
1.1	The Set of Humans	3
1.2	Well-Being as a Measurable Quantity	3
2	The Free Societal System	5
2.1	Formal Definition	5
3	Wealth and Its Relationship to Well-Being	7
3.1	The Wealth Function	7
3.2	The Logarithmic Relationship: $W \approx \ln A$	8
4	The Emergence of Inequality and the Failure of Free Markets	9
4.1	The Pareto Result in Free Systems	9
4.2	Aggregate Well-Being	10
4.3	The Instantaneous Well-Being Cost of Unequal Wealth Distribution . . .	11
5	Government and the Optimization of Collective Well-Being	15
5.1	The Governed Societal System	15
5.2	Moloch: A Formal Definition of Coordination Failure	17
5.2.1	The Concept	17
5.2.2	Formal Definition	17
6	Slaying Moloch: Policy Instruments for Well-Being Maximization	18
6.1	Acting on $\bar{A}(t)$: Environmental Destruction and Development	19
6.2	Acting on $\mathcal{I}(t)$: Redistribution and the $\ln \bar{A}(t) - \mathcal{I}(t)$ Tradeoff	19
6.3	Acting on $\frac{n(t)}{N}$: The Protection of Life	20
7	Structural Sources of Moloch in Contemporary Governance	21
7.1	The Nation-State Partition as a Source of Moloch	21
7.2	The Short-Cycle Democratic System as a Source of Moloch	23
7.3	What Makes a Good Governor for Collective Well-Being?	24
8	The Scope of the Welfare Set and AI Alignment	29
8.1	Why the Present Text Focuses on Human Well-Being	29
8.2	AI Deployments and Misalignment in the Human System	32
8.2.1	Three Structural AI Problems for Human Well-Being	36
9	Conclusion	37

1 The Individual and the Well-Being Function

1.1 The Set of Humans

Definition 1.1 (The Universal Set of Humans). Let $\mathcal{H}$ denote the **universal set of all humans who have ever existed or will ever exist**. We assume that $\mathcal{H}$ is **finite**, i.e., $|\mathcal{H}| = N$ for some $N \in \mathbb{N}$. This finiteness assumption rests on two grounds:

- (i) *Physical*: It may be unlikely that the universe will permit human life forever because of entropy and physics evolution coming from the cosmological constant;
- (ii) *Mathematical*: finiteness ensures that sums over $\mathcal{H}$ are well-defined without requiring measure-theoretic apparatus.

We index the elements of $\mathcal{H}$ as $\{h_1, h_2, \dots, h_N\}$.

Definition 1.2 (Lifespan). For each individual $h_i \in \mathcal{H}$, we define the **lifespan interval**

$$L_i := [b_i, d_i] \subset \mathbb{R},$$

where $b_i \in \mathbb{R}$ is the birth time and $d_i \in \mathbb{R}$ is the death time of h_i , with $b_i < d_i$. The quantity $d_i - b_i$ is the **duration of life** of individual h_i .

1.2 Well-Being as a Measurable Quantity

We now advance a foundational claim that well-being is a well-defined, real-valued, and continuous quantity. This claim is justified empirically and theoretically:

- (i) **Phenomenological experience**: At every waking moment in the present, conscious individuals feel an instantaneous quality of existence—a subjective well-being that smoothly varies from extreme suffering to extreme joy.
- (ii) **Evolutionary biology**: Natural selection predicts the existence of an internal reward and punishment system (a guidance signal) to steer complex organisms with behavioral flexibility toward fitness-enhancing actions and away from fitness-reducing ones.

Definition 1.3 (Existence of Well-Being). There exists a function

$$W : \mathcal{H} \times \mathbb{R} \rightarrow \mathbb{R}$$

such that for each $h_i \in \mathcal{H}$ and each time $t \in L_i$, the value $W(h_i, t)$ represents the **instantaneous well-being** of individual h_i at time t . We require:

- (i) $W(h_i, t)$ is defined only for $t \in L_i$ (one cannot experience well-being outside of one’s lifetime);
- (ii) $W(h_i, \cdot) : L_i \rightarrow \mathbb{R}$ is continuous ;

Remark 1.4 (Continuity vs. Discreteness). While time or conscious experience might be fundamentally discrete (e.g., at the Planck scale or via discrete neural frames), the discretization interval is so infinitesimal that modeling $W(h_i, \cdot)$ as a continuous function is mathematically robust and justified.

Remark 1.5 (The Alignment of Well-Being and Fitness). Evolved as a guidance signal, **pleasure** ($W > 0$) and **suffering** ($W < 0$) correlate with fitness-enhancing and fitness-reducing states, respectively. Let $F(h_i, t)$ denote the true instantaneous inclusive fitness in the ancestral environment. Natural selection tuned neural architectures such that, to a first approximation,

$$\text{sign}(W(h_i, t)) \approx \text{sign}(F(h_i, t)).$$

Thus, well-being acts as a *noisy, ordinally faithful proxy* for the real-valued fitness landscape. However, in modern environments, the proxy W can tragically misalign with the true objective F (e.g., via addictive substances).

Definition 1.6 (Lifetime Well-Being). The **lifetime well-being** of individual h_i is

$$\mathcal{W}(h_i) := \int_{b_i}^{d_i} W(h_i, t) dt.$$

This is well-defined by the continuity condition in Definition 1.3.

Observation 1.7 (The Natural Objective of an Individual). From the perspective of individual h_i , a natural objective is to maximize $\mathcal{W}(h_i)$ (and this might be the closest thing to the goal of life), i.e., to choose actions over the course of one’s life so as to maximize the integral of instantaneous well-being over the lifespan. This is a continuous-time analog of the standard utility-maximization framework in economics, with the crucial distinction that W is not a preference ordering over consumption bundles but a direct measure of experienced quality of life.

Remark 1.8 (On the Sign of $\mathcal{W}$ and the Fatalist Fallacy). The sign of $\mathcal{W}(h_i)$ seems to determine whether the life of h_i was, in aggregate, “worth living”. One might naively conclude that if $\mathcal{W}(h_i) < 0$ is possible, then ending one’s life could be justified. However this is essentially always a fatalist fallacy, for three reasons. **First**, the future integral $\int_t^{d_i} W(h_i, s) ds$ is almost never irrecoverably negative: the space of available corrective actions—changes in environment, relationships, habits, medical treatment—is vast, and human circumstances are not static. **Second**, W depends not only on objective conditions but on the cognitive lens through which they are perceived; a *positive perceptual*

orientation can shift $W(h_i, t)$ into positive territory even in objectively difficult situations, as documented extensively in positive psychology. **Third**, humans are sensitive to *trajectories*, not just levels: if $P(h_i, t)$ denotes one’s social or economic position (P will not be used later in this text and is uniquely present in this argument), the perception that $\dot{P}(h_i, t) > 0$ —that things are improving—can itself make $W(h_i, t) > 0$ even when P is low.

2 The Free Societal System

2.1 Formal Definition

We now formalize the concept of a *free society* (no law) as a set-theoretical object.

Definition 2.1 (The Universal Transaction Set). A transaction is an exchange of money (which is a quantification of economic value) for a real object or a service among two humans. Let $\mathcal{T}$ denote the **universal set of all transactions** that have ever occurred or will ever occur among members of $\mathcal{H}$. A **transaction** $\tau \in \mathcal{T}$ is a tuple

$$\tau = (h_i, h_j, v, t_\tau),$$

where $h_i, h_j \in \mathcal{H}$ are the parties involved (with h_i the sender and h_j the receiver), $t_\tau \in \mathbb{R}$ is the time of the transaction, and $v \in \mathbb{R}_{\geq 0}$ is the economic value exchanged. To be absolutely comprehensive, one would need to explicitly model the set of all existing currencies, fluctuating exchange rates, inflation dynamics over time, and debt mechanisms (which are really just time-delayed transactions with interest). However, specifically for the formal objectives of this text, tracking these currency and debt mechanics would provide essentially no additional conceptual relevance. We therefore fix our quantification of money (v) to an abstract, universal unit of account (a numéraire). We assume $\mathcal{T}$ is **finite** (for the same reasons that $\mathcal{H}$ is finite: physical constraints on the universe).

Remark 2.2 (On the Omission of Firms as Separate Entities). Although real societies contain firms and corporations that appear, from a legal standpoint, to be distinct entities, we do not model them explicitly in this framework. This is partly for simplicity, and partly because the wealth formally held by such firms is, in most cases, ultimately owned by underlying humans such as shareholders or other beneficial claimants. For the formal purposes of this text, it is therefore sufficient to treat firms as intermediary structures through which humans transact and hold claims, rather than as fundamental agents in the ontology of the model.

Definition 2.3 (The Free Societal System). A **free societal system** is a function

$$S : \mathbb{R} \rightarrow 2^{\mathcal{H}} \times 2^{\mathcal{T}}$$

that assigns to each time $t \in \mathbb{R}$ a pair

$$S(t) = (H(t), \text{Tr}(t)),$$

where:

- (i) $H(t) \subseteq \mathcal{H}$ is the set of **living individuals at time t** :

$$H(t) := \{h_i \in \mathcal{H} : t \in L_i\}$$

- (ii) $\text{Tr}(t) \subseteq \mathcal{T}$ is the set of **transactions occurring at time t** :

$$\text{Tr}(t) := \{\tau = (h_i, h_j, v, t_\tau) \in \mathcal{T} : t_\tau = t\}.$$

- (iii) $n(t) := |H(t)|$ is the **population function**, giving the number of living individuals at time t . Throughout the text, we write the set of living individuals as $H(t) = \{h_{k_1(t)}, h_{k_2(t)}, \dots, h_{k_{n(t)}(t)}\}$, where the indices $k_j(t)$ identify the specific humans alive at time t .

Remark 2.4 (Local vs. Integral Optimality of Transactions). In a free societal system, individuals initiate transactions of their own volition typically because doing so increases their instantaneous well-being W locally in time. They willingly exchange formal value v for an object or service that they perceive as holding strictly greater value to them in that precise moment. However, it is clear that in such a free society, these myopic, locally optimizing transactions are not necessarily optimal for the total lifetime well-being (the integral $\mathcal{W}$) of the two individuals involved, this is because human brains are wired to be more short term oriented for immediate survival, and also because we navigate the world with less information about the far future than the immediate present.

Remark 2.5. Observe that $H(t)$ and $\text{Tr}(t)$ are time slices of the universal sets $\mathcal{H}$ and $\mathcal{T}$, respectively. At any given time t , only a subset of all humans who will ever live are alive, and only a subset of all transactions that will ever occur are occurring. The universal sets $\mathcal{H}$ and $\mathcal{T}$ are *atemporal* (timeless) collections—they are the “God’s-eye view” of the entirety of human history and economic activity.

Definition 2.6 (Temporal Extent of a Society). The **temporal extent** of the societal system is the interval

$$T_S := \left[\min_{h_i \in \mathcal{H}} b_i, \max_{h_i \in \mathcal{H}} d_i \right],$$

i.e., from the birth of the first human to the death of the last human.

Remark 2.7 (Biological Boundaries). We intentionally omit non-pertinent (for the intent of this text) biological details regarding the precise evolutionary transition from non-

human primates to humans. For the purpose of this framework, we assume $\mathcal{H}$ is a sharply defined set, abstracting away from the gradualism of natural selection.

3 Wealth and Its Relationship to Well-Being

3.1 The Wealth Function

Definition 3.1 (Individual Wealth). The **wealth function** (or monetary function) is a map

$$A : \mathcal{H} \times \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$$

such that $A(h_i, t)$ represents the total monetary wealth of individual h_i at time $t \in L_i$.

Remark 3.2. The quantity A should be understood in *real-value* terms: it represents money insofar as money faithfully tracks the value of actual resources, goods, services, and productive assets. In other words, A is not merely the nominal number of currency units held by an individual; it is the amount of genuine economic value that those holdings command. For a living human, this effective resource access is never literally zero in the sense relevant to well-being: even an individual with no financial wealth may still have access to food, temporary shelter, public goods, social support, infrastructure, or at least the biological flow of resources required to remain alive. Thus A should be read as the real value effectively accessible to the individual, which may be extremely small but remains strictly positive during life. If a government prints and distributes additional dollars without increasing the real stock of valuable resources in society, this does not increase A : the same total value is simply spread across a larger number of nominal currency units, so the dollar becomes weaker. Also, environmental destruction and pollution can reduce the future value of A even if they do not reduce the number of dollars in circulation, because they destroy resources and forms of natural value—food, usable land, clean air, health, comfort, and other valuable conditions supplied by nature—that would otherwise contribute to the real wealth of individuals.

Remark 3.3. The wealth function A is related to, but distinct from, the transaction set $\text{Tr}(t)$. Informally, the wealth $A(h_i, t)$ of an individual evolves according to the cumulative net effect of all transactions involving that individual. We emphasize that activities such as labor (selling one’s time as a service) and investment (selling the service of capital provision) are included within the transaction set $\text{Tr}(t)$, they are within the scope of the definition.

3.2 The Logarithmic Relationship: $W \approx \ln A$

A very important question is whether or not $W(h_i, t)$ admits a set-theoretic decomposition into collectively exhaustive and not inter-inclusive components. There are ways to construct such a decomposition; for instance:

$$W(h_i, t) = f\left(\begin{array}{l} \text{Past Accomplishments}(h_i, t), \text{Purpose}(h_i, t), \text{Social Recognition}(h_i, t), \\ \text{Intimate Relationships}(h_i, t), \text{Health}(h_i, t), \text{Serenity}(h_i, t), \\ \text{Aesthetic Contemplation}(h_i, t) \end{array}\right).$$

But one might find that money often acts as a prerequisite or an accelerator for these sources of well-being. For instance, wealth can secure medical treatments, provide the free time necessary for serenity, or facilitate social accomplishment. There is therefore generally a nontrivial causal link from having more money to being more happy. In fact, a robust empirical finding in the economics of happiness is that subjective well-being scales with the *logarithm* of income or wealth (they are typically proportional, see 4.1). This has been documented extensively:

- (i) Stevenson and Wolfers (2008) [1] demonstrated a strong linear-log relationship between subjective well-being and income across countries and within countries, challenging the Easterlin Paradox.
- (ii) Kahneman and Deaton (2010) [2] showed that *life evaluation* (a cognitive assessment of one's life) rises steadily with $\ln(\text{income})$, while *emotional well-being* plateaus around \$75,000/year (in 2010 USD).
- (iii) Killingsworth, Kahneman, and Mellers (2023) [3] refined this picture in an adversarial collaboration, showing that for the majority of individuals, well-being continues to rise with log-income even beyond the \$75,000 threshold.

We therefore adopt the following empirical result.

Observation 3.4 (Logarithmic Well-Being). For all $h_i \in \mathcal{H}$ and $t \in L_i$ with $A(h_i, t) > 0$, the well-being function satisfies, to leading order,

$$W(h_i, t) \approx \ln A(h_i, t).$$

Remark 3.5. The logarithmic form $W \sim \ln A$ has deep implications:

- (i) *Diminishing marginal utility*: $\frac{\partial W}{\partial A} = \frac{1}{A}$, which is strictly decreasing in A . A dollar given to a poor person increases well-being more than a dollar given to a rich person.
- (ii) *Scale invariance*: well-being depends on *proportional* changes in wealth, not absolute changes. Doubling one's wealth from \$1,000 to \$2,000 yields the same well-being gain as doubling from \$1,000,000 to \$2,000,000.

4 The Emergence of Inequality and the Failure of Free Markets

4.1 The Pareto Result in Free Systems

We now quote a standard result from the literature on wealth dynamics: when wealth accumulation is approximately multiplicative, and when the population experiences turnover through inheritance, bankruptcy, firm creation, firm destruction, and death, the upper tail of the stationary wealth distribution is generically Pareto [4, 5, 7, 8]. In continuous time, the core intuition of multiplicative accumulation is that

$$\frac{dA}{dt} \propto A,$$

meaning that the rate at which wealth grows is approximately proportional to the amount of wealth already held.

This is a realistic and observed approximation in our societies for at least two reasons. First, money can be invested: if an individual possesses a larger stock of wealth A , that person can deploy a larger amount of capital into productive assets, businesses, financial markets, or interest-bearing instruments, so the expected absolute gain tends to scale with existing wealth. Second, a person who has already accumulated a substantial amount of money is generally more likely to acquire even more: wealth improves access to credit, education, networks, information, bargaining power, and risk-taking capacity. In that sense, larger wealth does not merely generate returns mechanically; it also increases the practical probability of capturing further economic opportunities.

Proposition 4.1 (Pareto Emergence in Free Wealth Dynamics). *In broad classes of decentralized wealth-accumulation models with multiplicative growth and population turnover, the stationary cross-sectional wealth distribution has a Pareto upper tail:*

$$\Pr(A > a) \sim a^{-\alpha}$$

for large a and some Pareto exponent $\alpha > 1$.

This is a result from the literature [4, 5, 7, 8], which we cite here without proof.

This proposition matters because it shows that strong inequality in free systems is not an extra postulate. It is itself a result: pareto-type concentration arises endogenously.

Remark 4.2 (Empirical Pareto Exponents). Empirically, the wealth distribution in most countries exhibits $\alpha \in [1.5, 3]$, with $\alpha \approx 2$ being typical.

4.2 Aggregate Well-Being

Definition 4.3 (Naked Collective Well-Being). The **naked collective well-being** of the societal system S is

$$\mathcal{W}_{\text{coll}}^{\Sigma} := \sum_{i=1}^N \int_{b_i}^{d_i} W(h_i, t) dt = \sum_{i=1}^N \mathcal{W}(h_i).$$

Equivalently, isolating the monetary component and using Observation 3.4:

$$(\mathcal{W}_{\text{coll}}^{\Sigma})^{(\text{monetary})} = \sum_{i=1}^N \int_{b_i}^{d_i} \ln A(h_i, t) dt.$$

Definition 4.4 (Collective Well-Being). The **collective well-being** of the societal system S is the normalized quantity

$$\mathcal{W}_{\text{coll}} := \frac{1}{N} \mathcal{W}_{\text{coll}}^{\Sigma} = \frac{1}{N} \sum_{i=1}^N \mathcal{W}(h_i).$$

Thus, $\mathcal{W}_{\text{coll}}$ is the average lifetime well-being of an individual member of the society. When comparing candidate societal trajectories with different populations, the normalization is taken with respect to the number of individuals realized in the trajectory under consideration.

Definition 4.5 (Instantaneous Collective Quantities). At a fixed time t , define the **naked instantaneous collective well-being**

$$w_{\Sigma}(t) := \sum_{h_i \in H(t)} W(h_i, t) \approx \sum_{h_i \in H(t)} \ln A(h_i, t),$$

and the **instantaneous collective well-being**

$$w(t) := \frac{1}{n(t)} w_{\Sigma}(t) \approx \frac{1}{n(t)} \sum_{h_i \in H(t)} \ln A(h_i, t).$$

Remark 4.6 (Integral Relation with the Aggregate Quantity). If we extend $W(h_i, t)$ to all $t \in \mathbb{R}$ by setting $W(h_i, t) = 0$ whenever $t \notin L_i$, then

$$\int_{\mathbb{R}} w_{\Sigma}(t) dt = \int_{\mathbb{R}} \sum_{h_i \in H(t)} W(h_i, t) dt = \sum_{i=1}^N \int_{b_i}^{d_i} W(h_i, t) dt = \mathcal{W}_{\text{coll}}^{\Sigma}.$$

Thus, the naked collective well-being of the whole societal trajectory is the time-integral of the naked instantaneous collective well-being. By contrast, the analogous identity does

not hold for the normalized quantity $\mathcal{W}_{\text{coll}}$, because

$$w(t) = \frac{1}{n(t)} w_{\Sigma}(t)$$

contains the time-dependent factor $n(t)$, whereas

$$\mathcal{W}_{\text{coll}} = \frac{1}{N} \mathcal{W}_{\text{coll}}^{\Sigma}$$

is normalized by the total number N of realized individuals in the trajectory.

Remark 4.7 (Why the Normalized Objective Is Preferred). The naked quantity $\mathcal{W}_{\text{coll}}^{\Sigma}$ is useful descriptively, because it measures the total amount of well-being realized by a society. However, as a normative objective it has a serious pathology: if the number of realized humans can vary across candidate societal trajectories, then it rewards population growth even when the additional individuals have only arbitrarily small positive well-being. In the limit, it can prefer a world with extremely many barely happy humans to a world with fewer but much happier humans, and can therefore justify sacrificing any particular individual for the sake of increasing headcount. By contrast, $\mathcal{W}_{\text{coll}}$ measures the average lifetime well-being of an individual member of the society. Since every observation of the world is made by an individual, and a society is nothing over and above the aggregation of individuals, $\mathcal{W}_{\text{coll}}$ is the more appropriate normative objective for a society ($\mathcal{W}_{\text{coll}}^{\Sigma}$ is naked because it forgot to wear its normalization for the individuals). A mathematically clearer way to state the comparison is that, between these two candidate social objectives, for every human h_i in $\mathcal{H}$, this human would individually prefer $\mathcal{W}_{\text{coll}}$ to $\mathcal{W}_{\text{coll}}^{\Sigma}$ by definition (higher $\mathcal{W}(h_i)$ in average).

4.3 The Instantaneous Well-Being Cost of Unequal Wealth Distribution

Having stated the Pareto result for free systems, we now prove the relevant local claim for market failure. Fix a time t , evaluate the society by the *instantaneous* collective well-being $w(t)$, and fix the total amount of money/wealth $A_{\text{total}}(t)$ at that instant. Under these assumptions, the wealth distribution that arises from unregulated free markets does *not* maximize collective well-being.

Theorem 4.8 (Free Markets Do Not Maximize Collective Well-Being). *Fix a time t . Let $n = n(t)$ denote the population at that time, and suppose the total wealth*

$$A_{\text{total}}(t) = \sum_{h_i \in H(t)} A(h_i, t)$$

is fixed. Under the logarithmic well-being assumption of Observation 3.4, the instant-

neous collective monetary well-being

$$w(t) = \frac{1}{n(t)} \sum_{h_i \in H(t)} \ln A(h_i, t)$$

is maximized, subject to the constraint $\sum_{h_i \in H(t)} A(h_i, t) = A_{\text{total}}(t)$, if and only if

$$A(h_i, t) = \frac{A_{\text{total}}(t)}{n(t)} \quad \text{for all } h_i \in H(t),$$

i.e., wealth is **perfectly equally distributed**. Any Pareto-distributed (or otherwise unequal) allocation yields strictly lower collective well-being.

Proof. Fix t and enumerate the living individuals as

$$H(t) = \{h_{k_1(t)}, h_{k_2(t)}, \dots, h_{k_{n(t)}(t)}\}.$$

Write

$$a_j := A(h_{k_j(t)}, t), \quad j = 1, \dots, n(t).$$

Then

$$w(t) = \frac{1}{n(t)} \sum_{j=1}^{n(t)} \ln a_j.$$

Using the identity

$$\frac{1}{n(t)} \sum_{j=1}^{n(t)} \ln a_j = \ln \left(\left(\prod_{j=1}^{n(t)} a_j \right)^{1/n(t)} \right),$$

we see that $w(t)$ is the logarithm of the geometric mean of the wealth levels $(a_1, \dots, a_{n(t)})$.

By the arithmetic-geometric mean inequality,

$$\left(\prod_{j=1}^{n(t)} a_j \right)^{1/n(t)} \leq \frac{1}{n(t)} \sum_{j=1}^{n(t)} a_j = \frac{A_{\text{total}}(t)}{n(t)}.$$

Since $\ln$ is increasing, this implies

$$w(t) \leq \ln \frac{A_{\text{total}}(t)}{n(t)}.$$

Equality in the arithmetic-geometric mean inequality holds if and only if all the a_j are equal, i.e.

$$a_j = \frac{A_{\text{total}}(t)}{n(t)} \quad \text{for all } j.$$

Equivalently,

$$A(h_i, t) = \frac{A_{\text{total}}(t)}{n(t)} \quad \text{for all } h_i \in H(t).$$

Thus the unique maximizer is the perfectly equal allocation. Any Pareto-distributed (or otherwise unequal) allocation has strictly positive inequality term and therefore strictly lower collective well-being. $\square$

Remark 4.9 (Decomposition of Instantaneous Collective Well-Being). Let

$$\bar{A}(t) := \frac{A_{\text{total}}(t)}{n(t)}$$

denote the **average resource per capita** at time t . In this context the instantaneous collective well-being admits the decomposition

$$w(t) = \ln \bar{A}(t) - \mathcal{I}(t),$$

where

$$\mathcal{I}(t) := \ln \left(\frac{\bar{A}(t)}{\left(\prod_{h_i \in H(t)} A(h_i, t) \right)^{1/n(t)}} \right) \geq 0.$$

The first term depends only on the total resources present in the society at time t , while the second depends only on the equality of the distribution: it is zero under perfect equality and increases as the geometric mean falls below the arithmetic mean. Intuitively, this is why $\mathcal{I}(t)$ quantifies inequality. If total wealth $A_{\text{total}}(t)$ is held fixed, then the arithmetic mean $\bar{A}(t)$ is unchanged by redistribution. By contrast, the geometric mean is sensitive to dispersion: making the distribution more unequal creates smaller and larger individual wealth levels, and the product is disproportionately reduced by the smaller terms. Hence the geometric mean falls, the gap between the arithmetic and geometric means widens, and $\mathcal{I}(t)$ increases. In that sense, $\mathcal{I}(t)$ measures how far the distribution is from perfect equality.

Proof. We have

$$w(t) = \frac{1}{n(t)} \sum_{h_i \in H(t)} \ln A(h_i, t) = \ln \left(\left(\prod_{h_i \in H(t)} A(h_i, t) \right)^{1/n(t)} \right).$$

Adding and subtracting $\ln(A_{\text{total}}(t)/n(t))$ yields

$$w(t) = \ln \frac{A_{\text{total}}(t)}{n(t)} - \ln \left(\frac{A_{\text{total}}(t)/n(t)}{\left(\prod_{h_i \in H(t)} A(h_i, t) \right)^{1/n(t)}} \right),$$

which is exactly the stated decomposition. The non-negativity of $\mathcal{I}(t)$ follows from the arithmetic-geometric mean inequality. $\square$

Lemma 4.10 (Intertemporal Decomposition of Collective Well-Being). *Under the logarithmic well-being assumption of Observation 3.4, the monetary component of collective well-being satisfies*

$$(\mathcal{W}_{\text{coll}})^{(\text{monetary})} = \int_{\mathbb{R}} \frac{n(t)}{N} \left(\ln \bar{A}(t) - \mathcal{I}(t) \right) dt.$$

Thus, in the logarithmic monetary model, collective well-being is obtained by integrating through time a term in which $n(t)/N$ is the proportion of the total realized population that is alive at time t , $\ln \bar{A}(t)$ is the per-capita resource term, and $\mathcal{I}(t)$ is the inequality penalty.

Proof. By the definition of normalized collective well-being together with the integral relation for the naked aggregate quantity,

$$(\mathcal{W}_{\text{coll}})^{(\text{monetary})} := \frac{1}{N} (\mathcal{W}_{\text{coll}}^{\Sigma})^{(\text{monetary})} = \frac{1}{N} \int_{\mathbb{R}} w_{\Sigma}(t) dt.$$

Since

$$w_{\Sigma}(t) = n(t) w(t),$$

Remark 4.9 gives

$$w_{\Sigma}(t) = n(t) \left(\ln \bar{A}(t) - \mathcal{I}(t) \right).$$

Substituting into the previous identity yields

$$(\mathcal{W}_{\text{coll}})^{(\text{monetary})} = \frac{1}{N} \int_{\mathbb{R}} n(t) \left(\ln \bar{A}(t) - \mathcal{I}(t) \right) dt = \int_{\mathbb{R}} \frac{n(t)}{N} \left(\ln \bar{A}(t) - \mathcal{I}(t) \right) dt,$$

which is the desired formula. □

Remark 4.11 (Relationship to the Classical Literature). The instantaneous decomposition $w(t) = \ln \bar{A}(t) - \mathcal{I}(t)$ of Remark 4.9 is closely related to the welfare-theoretic interpretation of the Atkinson inequality index [16] for the special case of logarithmic utility ($\epsilon = 1$ in Atkinson’s parameterization). However, the intertemporal formula of Lemma 4.10, which integrates this decomposition over the full temporal extent of the society with the demographic weight $n(t)/N$, does not appear in this unified form in the existing literature. Its value lies in providing a single expression from which the principal channels of governmental action—acting on per-capita resources ($\ln \bar{A}$), on distributional inequality ($\mathcal{I}$), and on the preservation of human life ($n(t)/N$)—can be read off directly, as we show in Section 6.

5 Government and the Optimization of Collective Well-Being

The need for an organ of coordination—a government—arises directly from the preceding analysis. A free society tends generically toward strong inequality, and it can also fail to coordinate globally so as to increase the total stock of valuable resources available to its members. Left to itself, it therefore generally does a poor job of maximizing collective well-being. From a political point of view, it is often reasonable to reduce well-being to its monetary component. The reason is that most major sources of well-being can be purchased, or at least strongly facilitated, by wealth; while the sources of well-being that cannot be purchased are, in most cases, not directly producible by government policy either. In that sense, political optimization operates primarily through the management, production, preservation, and distribution of real economic value. Accordingly, throughout the governmental analysis that follows, we treat collective well-being as identified with its monetary component:

$$\mathcal{W}_{\text{coll}} = (\mathcal{W}_{\text{coll}})^{(\text{monetary})}.$$

5.1 The Governed Societal System

Definition 5.1 (Government). A **government** for a free societal system S is a pair

$$G = (g, \Pi),$$

where:

- (i) $g \notin \mathcal{H}$ is an additional entity, called the **government**, which is allowed to hold wealth through a treasury function

$$A_{\text{gov}} : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0};$$

- (ii) Π is a **policy function** which, at each time t , takes as input the history of transactions before t and outputs a complete rearrangement of the transactions occurring at time t .

Formally, let

$$\mathcal{T}_G \subseteq (\mathcal{H} \cup \{g\}) \times (\mathcal{H} \cup \{g\}) \times \mathbb{R}_{\geq 0} \times \mathbb{R}$$

denote the extended transaction space in which the government may also participate, and define the transaction history before time t by

$$\text{Tr}_{<t}^{(G)} := \{\tau \in \mathcal{T}_G : t_\tau < t\}.$$

Then

$$\Pi : 2^{\mathcal{T}_G} \times \mathbb{R} \rightarrow 2^{\mathcal{T}_G}, \quad \Pi(\text{Tr}_{<t}^{(G)}, t) = \text{Tr}^{(G)}(t),$$

where every transaction in $\text{Tr}^{(G)}(t)$ has time-coordinate t . Thus the government may delete, preserve, modify, or create any transaction at time t .

Definition 5.2 (Governed Societal System). A **governed societal system** is a pair (S, G) , where S is a free societal system (Definition 2.3) and $G = (g, \Pi)$ is a government adjoined to it. At each time t , the transaction slice that actually occurs is no longer the free slice $\text{Tr}(t)$, but the rearranged slice

$$\text{Tr}^{(G)}(t) = \Pi(\text{Tr}_{<t}^{(G)}, t).$$

The human wealth trajectories, the governmental treasury A_{gov} , and hence the well-being trajectories are all determined by this governed transaction process. In that sense, a government is a coordination layer added on top of a free societal system, with the power to rewrite the entire transaction slice at each instant.

Definition 5.3 (The Objective of Government). The **normative role** of a government is to maximize collective well-being over the temporal extent of the society:

$$\mathcal{W}_{\text{coll}}^*(S) := \sup_{G \in \mathcal{G}} \mathcal{W}_{\text{coll}}(S, G) = \sup_{G \in \mathcal{G}} \frac{1}{N} \sum_{i=1}^N \int_{b_i}^{d_i} W^{(G)}(h_i, t) dt,$$

where $\mathcal{G}$ is the set of all feasible governments, $W^{(G)}(h_i, t)$ denotes the well-being function under the governed transaction process induced by G , and N is understood as the number of individuals realized under the government being evaluated. If the supremum is attained by some feasible government, we denote any such maximizing government by G^* . The naked quantity $\mathcal{W}_{\text{coll}}^\Sigma$ remains useful for measuring the total historical scale of well-being, but it is not the primary normative target.

Remark 5.4 (Biological Finiteness of $\mathcal{W}_{\text{coll}}^*(S)$). The use of a supremum does not imply that the achievable collective well-being could be infinite. Within the biological interpretation of well-being adopted in this text, it is overwhelmingly implausible that instantaneous human well-being is unbounded above. Human experience is implemented by a finite physical substrate: a finite number of cells, a finite number of neurons, finite firing rates, finite metabolic throughput, and finite biochemical state-space. We therefore adopt the biologically natural boundedness assumption that there exists a universal constant $B_{\text{bio}} < \infty$ such that

$$W^{(G)}(h_i, t) \leq B_{\text{bio}} \quad \text{for all feasible } G, h_i \in \mathcal{H}, t \in L_i.$$

Since $\mathcal{H}$ is finite and each lifespan interval $L_i = [b_i, d_i]$ has finite length, it follows that

for every feasible government G ,

$$\mathcal{W}_{\text{coll}}(S, G) \leq \frac{1}{N} \sum_{i=1}^N B_{\text{bio}}(d_i - b_i) \leq B_{\text{bio}} \max_{1 \leq i \leq N} (d_i - b_i) < \infty.$$

Hence the optimal achievable value $\mathcal{W}_{\text{coll}}^*(S)$ is finite. In other words, switching from $\arg \max$ to $\sup$ addresses the possible non-attainment of the optimum, not the possibility of infinite well-being.

5.2 Moloch: A Formal Definition of Coordination Failure

5.2.1 The Concept

The term *Moloch*, introduced in this context by Alexander (2014) [6] following Ginsberg’s poem *Howl*, refers to the destructive force of coordination failure—the phenomenon whereby individually rational actors produce collectively catastrophic outcomes. We now give Moloch a precise mathematical definition.

5.2.2 Formal Definition

Definition 5.5 (The Moloch Number). Let (S, G) be a governed societal system. The **Moloch number** of the system (S, G) is the real number

$$\mathfrak{M}(S, G) := \mathcal{W}_{\text{coll}}^*(S) - \mathcal{W}_{\text{coll}}(S, G) \geq 0.$$

Proposition 5.6 (Properties of the Moloch Number). *(i) $\mathfrak{M}(S, G) \geq 0$ for all (S, G) , with equality if and only if G attains the supremum $\mathcal{W}_{\text{coll}}^*(S)$. In particular, if a maximizing government G^* exists, then equality holds if and only if $G = G^*$ (modulo policies that yield the same $\mathcal{W}_{\text{coll}}$).*

(ii) $\mathfrak{M}$ has units of “well-being $\times$ time” (the same units as $\mathcal{W}_{\text{coll}}$ and as individual lifetime well-being).

*(iii) The **normalized Moloch number** is*

$$\hat{\mathfrak{M}}(S, G) := \frac{\mathfrak{M}(S, G)}{\mathcal{W}_{\text{coll}}^*(S)},$$

whenever $\mathcal{W}_{\text{coll}}^(S) > 0$. It measures the well-being loss as a fraction of the optimal achievable collective well-being. Moreover,*

$$\hat{\mathfrak{M}}(S, G) \in [0, 1] \quad \text{whenever } \mathcal{W}_{\text{coll}}(S, G) \geq 0,$$

while $\hat{\mathfrak{M}}(S, G) > 1$ whenever $\mathcal{W}_{\text{coll}}(S, G) < 0$. A society with $\hat{\mathfrak{M}} = 0$ has slain Moloch; a society with $\hat{\mathfrak{M}} \rightarrow 1$ loses nearly all of its positive achievable well-being to coordination failure; and a society with $\hat{\mathfrak{M}} > 1$ lies in the regime of net-negative realized collective well-being.

Remark 5.7. The Moloch number captures, in a single real number, the well-being cost of all failures of coordination, incentive misalignment, corruption, regulatory capture, rent-seeking, and suboptimal policy in the governed societal system. It is, in a precise sense, the “distance from utopia.”

6 Slaying Moloch: Policy Instruments for Well-Being Maximization

The space $\mathcal{G}$ of feasible governments is extremely vast. There exist innumerable possible rule-systems, tax schedules, subsidies, prohibitions, institutional architectures, and enforcement procedures that could be imposed on a society, each with its own impact on collective well-being. Fortunately, the intertemporal decomposition of Lemma 4.10 gives strong conceptual guidance, as the three terms of the integrand directly identify the three principal channels through which a government can increase collective well-being:

$$\mathcal{W}_{\text{coll}} = \int_{\mathbb{R}} \frac{n(t)}{N} \left(\ln \bar{A}(t) - \mathcal{I}(t) \right) dt.$$

A naive reaction to the vastness of $\mathcal{G}$ would be to redesign the entire transaction structure of society from scratch. But a governmental system that is too random, too complicated, or too far from the baseline free society is conceptually disfavored for two structural reasons. First, the farther a government moves from the free system, the more rearrangement it must perform, and rearrangement is itself costly. Administration, information-gathering, monitoring, enforcement, compliance burdens, and coercive frictions all consume real resources and thereby depress $\bar{A}(t)$. Second, the basic transaction system of a free society already contains a powerful resource-generating mechanism: humans are rewarded for producing objects and services that other humans value highly enough to buy. Voluntary exchange therefore tends structurally to push the resource term upward.

For these reasons, the conceptually natural task of government is not to reorganize all transactions, but to introduce the smallest possible set of targeted rearrangements that correct the characteristic weaknesses of the free system. This makes the optimization problem far more tractable. The formula above shows that there are three principal angles of attack for reducing Moloch: one can act on $\bar{A}(t)$, on $\mathcal{I}(t)$, and on the temporal population factor $n(t)/N$.

6.1 Acting on $\bar{A}(t)$: Environmental Destruction and Development

Because a free society is constituted by transactions, the direct way to act on $\bar{A}(t)$ is to act on transactions themselves. A rise in $\bar{A}(t)$ corresponds to an improvement of the environment in the broad sense: not only natural ecosystems, but any surrounding resource substrate that supports future production and well-being, including urban infrastructure, social trust, institutional competence, public cleanliness, human health, and knowledge capital. Likewise, a fall in $\bar{A}(t)$ corresponds to the destruction or depletion of that environment.

Each transaction $\tau = (h_i, h_j, v, t_\tau)$ admits a natural set-theoretic decomposition of its resource effects across three classes: the seller h_i , the client h_j , and the rest of humanity $H(t) \setminus \{h_i, h_j\}$. Resource destruction can in principle fall on any of these three. It is comparatively rare that the seller destroys, in the same act, the very resource base from which the seller profits. It is much more common that the damage falls on third parties, as with air pollution, soil degradation, or the destruction of natural and urban ecosystems. It is also common that the damage falls on the future of the client: because humans are structurally myopic (see Remark 2.4), they sometimes enter transactions that produce short-term pleasure while weakening long-term wealth, health, or productive capacity, as with addictive drugs, alcohol, and other forms of exploitative consumption.

The direct policy tools are therefore conceptually natural. Transactions that destroy the environment should be discouraged by forced governmental transactions (taxes). Transactions that develop the environment, preserve resource-rich ecosystems, or strengthen the client's long-term productive capacity should be encouraged by forced transactions (public funding). In short, the natural way to raise $\bar{A}(t)$ is to tax resource destruction and to finance resource creation.

6.2 Acting on $\mathcal{I}(t)$: Redistribution and the $\ln \bar{A}(t) - \mathcal{I}(t)$ Tradeoff

The inequality term $\mathcal{I}(t)$ emerges structurally from the Pareto dynamics of free systems. If the objective is to lower $\mathcal{I}(t)$ directly, the conceptually natural move is forced redistribution. In set-theoretic terms, for any poverty threshold $a \in \mathbb{R}_{\geq 0}$, let

$$P(t, a) := \{h_i \in H(t) : A(h_i, t) < a\}$$

denote the subset of humans whose wealth lies below a at time t . Then the resources used to finance transfers toward $P(t, a)$ must structurally be drawn from its complement $H(t) \setminus P(t, a)$. Progressive taxation of wealth, taxation of the transactions of wealthy individuals, and transfers such as a universal basic income therefore arise as natural candidate instruments for lowering $\mathcal{I}(t)$.

Among these tools, a tax on wealth itself is especially natural. The reason is that

$\mathcal{I}(t)$ is defined from the cross-sectional distribution of wealth stocks, not from the flow of transactions alone. A tax indexed directly to large wealth holdings therefore acts more immediately on the quantity one is trying to compress than a tax that touches only individual exchanges.

Yet the problem is not one-dimensional. The same taxes that reduce inequality can also weaken the incentives, freedom, and capital accumulation that help generate resources in the first place. As taxation on wealthy individuals becomes more severe, it can discourage productive effort, investment, and entrepreneurial risk-taking, thereby lowering $\bar{A}(t)$. The government should therefore not minimize $\mathcal{I}(t)$ in isolation. Its optimization problem is to manage the tradeoff inside the integrand itself, namely to maximize through time the quantity

$$\ln \bar{A}(t) - \mathcal{I}(t),$$

and hence the full intertemporal objective

$$\int_{\mathbb{R}} \frac{n(t)}{N} \left(\ln \bar{A}(t) - \mathcal{I}(t) \right) dt.$$

The conceptually natural redistributive policy is therefore not maximal redistribution, but mathematically optimized redistribution with collected data from the real world to accurately model $\ln \bar{A}(t) - \mathcal{I}(t)$ with varying taxation and UBI thresholds.

6.3 Acting on $\frac{n(t)}{N}$: The Protection of Life

The final factor in the integral is the temporal population weight $n(t)/N$, the proportion of all realized humans who are alive at time t . Assuming, as one should in a well-governed society, that instantaneous collective well-being is positive, increasing this factor is desirable. The most direct way to do so is not to lose existing human lives which would lower $n(t)$ and leave N fixed. Protecting the lives of the humans already in the society therefore becomes a natural governmental objective: reducing preventable death, war, disease, infrastructure failure, and extreme violence all act in the correct direction.

For the same reason, one also sees that, from the societal point of view, it is desirable to develop longevity technologies. If humans who are already part of the realized population can remain alive longer while preserving positive well-being, then this raises $n(t)$ over a longer interval without automatically enlarging the denominator through the creation of new individuals. This also matches ordinary human preference: in a good society, each human would typically wish to live longer in order to enjoy more of life, rather than to have his or her lifespan cut short despite life remaining worth living.

By contrast, simply adding more humans is not automatically desirable under the normalized objective $\mathcal{W}_{\text{coll}}$. New humans also enlarge the denominator N , and can strain

the resource and inequality terms if the society is not improved correspondingly. The addition of a new human is therefore justified only if that addition makes the society better on average, or at least does not degrade the average collective well-being over the full temporal horizon.

This conclusion marks an important contrast with the naked aggregate objective $\mathcal{W}_{\text{coll}}^\Sigma$. If the naked sum were the primary normative target, one would obtain a strong pressure to add as many humans as possible as fast as possible, even if doing so imposed severe losses on already existing humans. Under the normalized objective $\mathcal{W}_{\text{coll}}$, that conclusion does not follow. For any human in the society, the normalized objective is therefore the conceptually desired quantity.

7 Structural Sources of Moloch in Contemporary Governance

We now analyze two dominant features of the current global political architecture—the plurality of sovereign nations and the short-cycle democratic system—as *structural generators of Moloch*, and propose a formal alternative grounded in the maximization of the normalized collective well-being $\mathcal{W}_{\text{coll}}$.

7.1 The Nation-State Partition as a Source of Moloch

Definition 7.1 (The Nation-State Partition). Let $\{N_1, N_2, \dots, N_m\}$ be the set of sovereign nations. At each time t , each nation determines a citizenship subset

$$N_j(t) \subseteq H(t),$$

and we assume only that these subsets cover the living population:

$$H(t) = \bigcup_{j=1}^m N_j(t).$$

They need not be pairwise disjoint; in particular, individuals with multiple nationalities may belong to several $N_j(t)$ simultaneously. For each living human $h_i \in H(t)$, let

$$c_i(t) := |\{j \in \{1, \dots, m\} : h_i \in N_j(t)\}|$$

denote the number of nations to which h_i belongs at time t . We then define the **effective size** of nation j by

$$\tilde{N}_j(t) := \sum_{h_i \in N_j(t)} \frac{1}{c_i(t)}.$$

By construction,

$$\sum_{j=1}^m \tilde{N}_j(t) = n(t),$$

so this weighting counts multiply affiliated individuals only once in aggregate.

Each nation N_j is governed by a government G_j that optimizes a **local** objective:

$$\mathcal{W}_{\text{coll},j}^* := \sup_{G_j \in \mathcal{G}_j} \mathcal{W}_{\text{coll}}(N_j, G_j),$$

i.e., each nation evaluates governments only by the collective well-being of its own citizens.

If the local supremum is attained, we denote any attaining government by G_j^* .

Observation 7.2 (The Structural Cost of Nation-States). The population-weighted combination of nationally optimized governments is structurally suboptimal for humanity as a whole. In symbols,

$$\sum_{j=1}^m \frac{\tilde{N}_j}{N} \mathcal{W}_{\text{coll},j}^* < \mathcal{W}_{\text{coll}}^*(S),$$

where $\tilde{N}_j$ is the effective size of nation j over the relevant comparison, normalized so that $N = \sum_{j=1}^m \tilde{N}_j$, and $\mathcal{W}_{\text{coll}}^*(S)$ is the globally optimal achievable value for the full societal system. The gap is precisely the **inter-national component** of $\mathfrak{M}$:

$$\mathfrak{M}_{\text{nations}} := \mathcal{W}_{\text{coll}}^*(S) - \sum_{j=1}^m \frac{\tilde{N}_j}{N} \mathcal{W}_{\text{coll},j}^*.$$

To see why this is a structural truth, let $\bar{A}_j(t)$ denote the average resources per capita within nation j at time t . Suppose nation j taxes transactions that destroy the environment. This policy is globally desirable because environmental destruction lowers the global resource term $\ln \bar{A}(t)$. However, in the absence of a world government, the individuals producing the polluting goods can simply relocate to another nation k where they are still allowed to pollute. From the narrow perspective of nation k , this inflow can raise $\ln \bar{A}_k(t)$ by attracting production, employment, and capital. Yet from the perspective of humanity, the same relocation still depresses the global term $\ln \bar{A}(t)$ because the environmental damage has merely been displaced rather than eliminated.

Exactly the same structure appears for redistribution. Suppose nation i imposes taxes designed to lower its inequality term. Rich individuals can then migrate, transfer capital, or reroute transactions toward nation j , where lower taxes make accumulation more attractive. For nation i , this flight lowers $\ln \bar{A}_i(t)$ by removing productive capital and wealthy taxpayers. For nation j , the same inflow can raise $\ln \bar{A}_j(t)$. But globally the result is perverse: the world distribution remains more unequal than it otherwise would, so the global inequality penalty stays too high.

The division into nations also makes mercantilism structurally tempting. A nation

may try to improve its own position not by increasing the total resources of humanity, but by taking resources from another nation through conquest or geopolitical domination. For the winning nation i , such appropriation can indeed increase $\ln \bar{A}_i(t)$ locally if resources are successfully transferred inward. But at the global level, war and coercive extraction do not create wealth: they destroy infrastructure, consume capital, kill humans, interrupt production, and redirect enormous resources toward violence rather than value creation. Thus even when mercantilism appears nation-rational, the global term $\ln \bar{A}(t)$ has only decreased.

In short, nation-level optimization systematically converts globally desirable policies into locally disadvantageous ones, and this is exactly why the existence of multiple uncoordinated sovereign governments generates an irreducible Moloch term. These three mechanisms—pollution leakage, redistribution-undermining tax competition, and mercantilist appropriation through war—are not merely theoretical deductions; they are all extensively observed in the real world.

Remark 7.3. Although uncoordinated sovereign nations are structurally suboptimal relative to good global coordination, they can still be preferable to a single very bad centralized global government. Institutional plurality reduces the probability that all humans are governed by the same defective policy regime at once. If one nation is badly governed, other nations may remain better governed, and the existence of different political experiments allows governments to learn from one another’s successes and failures over time.

Plurality also creates a disciplining mechanism against extremely bad local governance. If a government G_i in nation i adopts policies that severely reduce the national resource term $\ln \bar{A}_i(t)$, wealthy individuals, firms, and mobile capital may flee toward better-governed jurisdictions. This exit weakens the bad government by reducing its tax base, investment base, and productive capacity. Thus, even though capital flight can undermine globally desirable redistribution in the uncoordinated case, it can also serve as a check on governments whose policies catastrophically destroy local wealth.

The conclusion is therefore not that a single centralized world government is automatically preferable to a world of independent nations. A bad global government would create a single point of failure for humanity as a whole. The claim is instead comparative: a good global government, or a sufficiently good system of global coordination, would achieve a higher $\mathcal{W}_{\text{coll}}$ than independent nations optimizing locally without coordination, especially for problems whose externalities are global in scale.

7.2 The Short-Cycle Democratic System as a Source of Moloch

Definition 7.4 (The τ -Democratic System). A **τ -democratic system** is a governance mechanism in which the government $G(t)$ is selected by majority vote of the electorate

$E(t) \subseteq H(t)$ at intervals of duration $\tau > 0$ (typically $\tau = 4$ or 5 years). Let $V : E(t) \rightarrow \mathcal{G}$ denote the voting function. The elected government satisfies

$$G_{\text{elected}} = \arg \max_{G \in \mathcal{G}} |\{h_i \in E(t) : V(h_i) = G\}|.$$

Remark 7.5 (Structural Deficiencies of τ -Democracy). The τ -democratic system introduces at least three sources of Moloch:

- (i) **Temporal myopia.** Governmental rotation has an important anti-tyrannical function: it limits the duration of bad rule and helps prevent stable dictatorships. However, it also gives the elected government G_{elected} an incentive horizon of only τ years. Policies whose benefits materialize on timescales $\gg \tau$ (climate policy, infrastructure, education, longevity research) are systematically underweighted. Thus, even though rotation is protective against very bad governments, it remains structurally suboptimal for maximizing $\mathcal{W}_{\text{coll}}$ because it weakens long-term incentives. Formally, the effective objective becomes short term, for example:

$$\tilde{\mathcal{W}} \sim \int_t^{t+\tau} \frac{1}{n(s)} \sum_{h_i \in H(s)} W^{(G)}(h_i, s) ds \neq \mathcal{W}_{\text{coll}},$$

discarding the vast majority of the temporal extent of the society.

- (ii) **Competence-blind selection.** The voting function V aggregates preferences without any filter on the voter's understanding of governance, economics, or well-being theory. The modal voter optimizes based on incomplete information, cognitive biases, and short-term self-interest, not $\mathcal{W}_{\text{coll}}$.
- (iii) **Adversarial dynamics.** The competitive structure of elections rewards *persuasion* (rhetoric, charisma, media manipulation, wealth for marketing) over *the personal traits that make an individual maximize collective well-being*, further misaligning the selected leaders from G^* .

7.3 What Makes a Good Governor for Collective Well-Being?

The failure of τ -democracy to reliably select good governors raises a prior question: how should one measure the *efficiency* of a governor in the present framework? In the formalism of Section 5, a government is the pair $G = (g, \Pi)$, and the decisive object is the policy function Π that rearranges the transaction slice at each time. In ordinary language, the governor is the human decision-maker, or set of decision-makers, entrusted with choosing that policy function.

Definition 7.6 (Governor Efficiency). Let G be a government for the societal system S .

The **efficiency** of the governor associated with G is

$$\mathcal{E}(G) := \frac{1}{\mathfrak{M}(S, G)},$$

with the convention that $\mathcal{E}(G) = +\infty$ whenever $\mathfrak{M}(S, G) = 0$. Thus, a more efficient governor is one with a smaller Moloch number, and a perfect governor—one attaining the optimum whenever such attainment is possible—has infinite efficiency.

Definition 7.7 (Four Desiderata for Governor Selection). The present framework suggests four primary desiderata for selecting governors:

- **Wisdom:** correct understanding of the well-being function W and of the real determinants of human flourishing, rather than the optimization of a convenient but false proxy;
- **Conscientiousness:** the diligence, reliability, discipline, and long-term consistency needed to execute difficult policy over time;
- **Intelligence:** the capacity to model complex systems, anticipate indirect effects, process evidence, and infer which concrete policies follow from the objective;
- **Altruism:** the motivation to implement policy for the sake of other humans rather than for personal power, wealth, status, or tribal advantage.

Proposition 7.8 (A Necessary and Sufficient Decision Profile for High Governor Efficiency). *Let $h_i \in \mathcal{H}$ be a candidate governor. In the present framework, h_i has the conceptual profile of a highly efficient governor if and only if the following three conditions are jointly satisfied:*

- (i) h_i **understands** what actually increases collective well-being, i.e., understands which feasible policies move society toward maximizing $\mathcal{W}_{\text{coll}}$;
- (ii) h_i **chooses** those policies for the sake of society, rather than replacing them with policies that serve private, factional, tribal, or short-term electoral objectives;
- (iii) h_i **implements and sustains** those policies in the real institutional world through time.

Equivalently, the logical decomposition of governor efficiency is

$$\begin{aligned} \text{understand what increases } \mathcal{W}_{\text{coll}} &\iff \text{wisdom} + \text{intelligence}, \\ \text{choose those policies for society} &\iff \text{altruism}, \\ \text{implement and sustain those policies} &\iff \text{conscientiousness}. \end{aligned}$$

Hence, at the level of conceptual structure,

$$\text{high governor efficiency} \iff (\text{wisdom} \wedge \text{intelligence} \wedge \text{altruism} \wedge \text{conscientiousness}).$$

In particular, governor efficiency is logically increasing in the coordinated presence of these four traits, because each trait removes a distinct bottleneck in the chain from correct understanding to policy selection to durable execution.

Proof. The definition of efficiency makes clear that a governor is better exactly insofar as the induced policy path brings realized society closer to the optimum $\mathcal{W}_{\text{coll}}^*(S)$ and therefore lowers $\mathfrak{M}(S, G)$. Conceptually, this requires a complete decision chain: the governor must know what should be done, must actually select that course of action for the population, and must carry it into sustained institutional reality. If any link in that chain fails, efficiency logically fails with it.

The first link—understanding what increases collective well-being—requires both wisdom and intelligence. Wisdom is needed because the governor must identify the true target, namely human well-being rather than an attractive but false proxy. Intelligence is needed because, in a complex society, the correct target does not mechanically reveal the correct policy; one must infer concrete policy under uncertainty, strategic behavior, indirect effects, and institutional constraints. Thus the understanding stage decomposes into the coordinated action of wisdom and intelligence.

The second link—choosing the identified policy for society—is the alignment stage. Once the governor understands which policies best serve collective well-being, the remaining question is whether those policies are in fact selected, rather than replaced by policies serving the governor’s own power, wealth, status, ideology, or in-group. That selection problem is exactly what altruism resolves by definition.

The third link—implementing and sustaining the chosen policy through time—is the execution stage. Even correct and benevolent policy has little effect if it is not carried through difficult administrative, legal, and institutional work. Conscientiousness is therefore the trait that closes the chain by turning correct policy into durable practice.

Conversely, if these three stages are all present, then the governor logically possesses the full conceptual profile required to push policy toward the feasible maximization of $\mathcal{W}_{\text{coll}}$. This is exactly what it means, in the present framework, to be a highly efficient governor. The proposition follows. $\square$

Remark 7.9 (Why Wisdom and Intelligence Should Not Be Collapsed). It is tempting to compress wisdom and intelligence into a single trait, since both appear in the “understanding” stage above. But empirical work in psychology supports the claim that intelligence and wisdom are related but distinct, and in particular that intelligence is not a sufficient condition for wisdom [10]. A person may therefore be highly capable at modeling systems,

strategy, economics, law, or engineering, yet never have thought seriously enough about well-being itself, and may therefore optimize a convenient but mistaken proxy rather than genuine human flourishing. Conversely, wisdom without intelligence leaves the governor with reduced adaptive and fine-grained interpretive capacity: even if the principles of well-being are known, the governor may still fail to correctly understand particular situations, anticipate hidden interactions, or update appropriately under changing social conditions. In a modern society, long causal chains, feedback loops, strategic adaptation, and institutional complexity require substantial inferential power. Wisdom sets the target; intelligence is the tool needed to trace the conceptual path toward it.

Remark 7.10 (What Fails When One Trait Is Missing). The importance of the four traits becomes clearer by examining the four cases in which three are present but one is missing.

- (i) **Wise, intelligent, and conscientious, but not altruistic.** Such a governor may understand collective well-being perfectly and may be fully capable of implementing coherent policy, yet still choose to use that ability for private enrichment, dynastic advantage, ideological domination, or the flourishing of a narrow in-group. The problem here is not incompetence but misaligned motivation.
- (ii) **Wise, intelligent, and altruistic, but not conscientious.** Such a governor may genuinely understand what would help society and may sincerely wish to implement it, yet fail through inconsistency, procrastination, disorganization, weak follow-through, or inability to sustain difficult institutional work over long horizons. Good intentions and insight do not suffice if they are not translated into stable action.
- (iii) **Wise, conscientious, and altruistic, but not intelligent.** Such a governor may know the moral target and may work hard in service of it, yet still be unable to model complex side effects, strategic adaptation, feedback loops, budget constraints, or institutional second-order effects. In a modern society, benevolence without sufficient cognitive capacity can still produce systematically bad policy.
- (iv) **Intelligent, conscientious, and altruistic, but not wise.** This is perhaps the most deceptive failure mode. Such a governor can think hard, work hard, and genuinely care about others, yet still optimize the wrong proxy for well-being because the governor has not actually understood what makes human life go well. Surprisingly, this case is very real: intelligent people do not automatically redirect their intelligence toward understanding well-being itself. One can be brilliant at strategy, economics, law, engineering, or rhetoric and still be confused about the true structure of human well-being.

Remark 7.11 (The Filtering Problem Is Scientific). Once the target profile is stated clearly, the next question is no longer a vague philosophical one. It becomes a scientific

problem of measurement and prediction: how can we design the best tests for wisdom, conscientiousness, intelligence, and altruism, so as to filter for governors whose policies actually increase collective well-being?

This is exactly the kind of problem on which one would want the best available scientists. Psychometrics, behavioral economics, experimental psychology, neuroscience, political science, institutional design, and longitudinal social-science methods are all relevant. The natural conclusion is *not* that the best scientists should simply be placed in power. Scientific excellence and governor excellence are not the same property. Rather, the better institutional plan is to use the best scientists to design, validate, and continuously improve the filtering mechanisms that determine who should hold power.

At least at a first approximation, measuring conscientiousness, intelligence, and wisdom is not an alien problem for modern science. There already exist mature psychometric traditions for intelligence and personality, and substantial empirical work on the measurement of wisdom as a psychological construct [9, 10]. These measurements are imperfect, but they are well within the general reach of contemporary scientific method. The difficulty is not that the relevant traits are in principle beyond observation, but that they must be measured carefully enough to predict actual governing performance rather than merely laboratory performance.

The most interestingly difficult case is altruism. Once a society openly declares that altruism is a selection criterion for power, candidates acquire incentives to *perform* altruism rather than possess it. This is a special instance of the general measurement pathology captured by Campbell’s law: the more strongly a social indicator is used for decision-making, the more it becomes vulnerable to corruption and strategic distortion [11]. Related empirical work on socially desirable self-presentation likewise suggests that measuring prosocial motivation is especially vulnerable to contamination by signaling, impression management, and strategic self-report [12]. For that reason, the scientific problem is not merely to measure stated altruism, but to detect hidden preference, strategic deception, and counterfeit prosociality across time and circumstance.

This point connects directly to AI alignment. In the AI case as in the human case, the central difficulty is not only to evaluate overt behavior, but to determine whether an agent is producing genuinely aligned behavior or merely an aligned-looking performance. Recent work on deceptive behavior in language models underscores exactly this concern [13]. More generally, the ability to rationally distinguish sincere optimization from strategic mimicry—in human brains as in artificial ones—may be one of the most important scientific problems for the long-run protection of well-being.

It is worth noting that such a scientific filtering system is substantially more feasible today than at any previous point in human history. The maturity of modern psychometrics, neuroscience, and behavioral science, combined with the rapidly growing capacity of artificial intelligence to process longitudinal behavioral data and detect subtle patterns of

deception, makes the design of robust governor-selection mechanisms a realistic scientific undertaking rather than a speculative one.

8 The Scope of the Welfare Set and AI Alignment

8.1 Why the Present Text Focuses on Human Well-Being

One can describe the issue more abstractly by allowing the aggregation domain itself to vary.

Definition 8.1 (Candidate Welfare Set). Let $\mathcal{X}$ be any finite set such that $\mathcal{H} \subseteq \mathcal{X}$ and such that each element $x \in \mathcal{X}$ is a bearer of welfare with lifespan interval L_x and instantaneous welfare function

$$W_{\mathcal{X}} : \mathcal{X} \times \mathbb{R} \rightarrow \mathbb{R}.$$

We then define the normalized collective welfare over $\mathcal{X}$ by

$$\mathcal{W}_{\text{coll}}[\mathcal{X}] := \frac{1}{|\mathcal{X}|} \sum_{x \in \mathcal{X}} \int_{L_x} W_{\mathcal{X}}(x, t) dt.$$

If, in addition, there is an economically meaningful wealth map

$$A_{\mathcal{X}} : \mathcal{X} \times \mathbb{R} \rightarrow \mathbb{R}_{\geq 0},$$

then for the living slice

$$\mathcal{X}(t) := \{x \in \mathcal{X} : t \in L_x\}$$

one may define the associated per-capita resource term

$$\bar{A}_{\mathcal{X}}(t) := \frac{1}{|\mathcal{X}(t)|} \sum_{x \in \mathcal{X}(t)} A_{\mathcal{X}}(x, t)$$

and the corresponding inequality term

$$\mathcal{I}_{\mathcal{X}}(t) := \ln \bar{A}_{\mathcal{X}}(t) - \frac{1}{|\mathcal{X}(t)|} \sum_{x \in \mathcal{X}(t)} \ln A_{\mathcal{X}}(x, t).$$

The standing choice in the present text is the special case $\mathcal{X} = \mathcal{H}$, in which

$$\mathcal{W}_{\text{coll}}[\mathcal{H}] = \mathcal{W}_{\text{coll}}.$$

Definition 8.2 (Two Natural Enlargements of $\mathcal{H}$). Let $\mathcal{Y}$ denote the set of non-human animals, and let $\mathcal{M}$ denote the set of artificial intelligence systems. Then two natural

strict enlargements of $\mathcal{H}$ are

$$\mathcal{X}_{\text{HA}} := \mathcal{H} \cup \mathcal{Y}, \quad \mathcal{X}_{\text{HAI}} := \mathcal{H} \cup \mathcal{Y} \cup \mathcal{M}.$$

Remark 8.3 (Why $\mathcal{H}$ Is a Natural Choice in the Present Framework). Among the candidate welfare sets just defined, the human set $\mathcal{H}$ has four especially natural properties:

- (i) **First-person naturalness.** Relative to the present discourse, $\mathcal{H}$ is the smallest candidate welfare set that already contains the author, the reader, and their species that is inherently social. Or else some AI reader that is anyway, one hopes, aligned with human welfare. For that reason, $\mathcal{W}_{\text{coll}}[\mathcal{H}]$ is directly connected to what individually helps the beings situated inside the argument.
- (ii) **Comparability without extra weights.** The restriction of welfare to $\mathcal{H}$ is comparatively homogeneous: humans are biologically and cognitively similar enough that it is at least plausible to compare the values $W_{\mathcal{H}}(h_i, t)$ on a common real-valued scale without first introducing an auxiliary weighting map on the domain.
- (iii) **Resource centrality.** The economic apparatus already developed in the text is human-indexed: the wealth function A is defined on $\mathcal{H} \times \mathbb{R}$, the transaction set $\mathcal{T}$ records human exchanges, and the principal producers and controllers of present-day institutions and resources are humans. Thus the monetary part of the framework fits $\mathcal{H}$ especially naturally.
- (iv) **Relative non-customizability.** For $h_i \in \mathcal{H}$, the welfare function is largely an evolved target fixed by natural selection rather than a designer-chosen reward channel. This makes the objective comparatively stable.

For these reasons, $\mathcal{W}_{\text{coll}}[\mathcal{H}]$ is not the only coherent welfare objective, but it is the most natural one for the purposes of the present text and society.

Remark 8.4 (The Enlargement $\mathcal{X}_{\text{HA}} = \mathcal{H} \cup \mathcal{Y}$). The enlarged objective $\mathcal{W}_{\text{coll}}[\mathcal{X}_{\text{HA}}]$ remains morally and conceptually sensible. Yet relative to $\mathcal{W}_{\text{coll}}[\mathcal{H}]$, three of the four properties above become weaker.

First, because $\mathcal{H} \subsetneq \mathcal{X}_{\text{HA}}$, the normalization is now taken over a larger set of welfare-bearers, so the objective becomes less directly first-personal for existing humans: the same total stock of resources must now be spread across humans and animals, therefore partially sacrificing the average quantity of monetary value available for the reader or me. Second, welfare comparison becomes substantially harder. Once one moves from $\mathcal{H}$ to $\mathcal{X}_{\text{HA}}$, the raw codomain $\mathbb{R}$ is no longer obviously enough by itself. A weighted aggregate of the form

$$\mathcal{W}_{\text{coll}}^{\omega}[\mathcal{X}_{\text{HA}}] := \frac{1}{\sum_{x \in \mathcal{X}_{\text{HA}}} \omega(x)} \sum_{x \in \mathcal{X}_{\text{HA}}} \omega(x) \int_{Lx} W_{\mathcal{X}_{\text{HA}}}(x, t) dt$$

begins to look natural, where $\omega : \mathcal{X}_{\text{HA}} \rightarrow \mathbb{R}_{>0}$ is a correction map. But the problem of choosing ω so that it tracks the real structure of welfare across species is not something that can be cleanly solved by conceptual analysis alone.

Third, the resource-centrality property weakens. Most members of $\mathcal{Y}$ bear welfare but produce little or no transferable economic wealth in the human monetary sense. If one nevertheless extends the economic approximation to $\mathcal{X}_{\text{HA}}$, then the analogue $\mathcal{I}_{\mathcal{X}_{\text{HA}}}(t)$ of the inequality term becomes vastly larger, since many more sentient beings now sit near the bottom of the resource distribution. A government optimizing over $\mathcal{X}_{\text{HA}}$ would therefore be pushed toward much stronger provisioning and redistribution, including animal-directed support structurally analogous to a kind of UBI.

The fourth property, however, largely remains: welfare on $\mathcal{Y}$, like welfare on $\mathcal{H}$, is still an evolved target rather than a freely programmable reward channel.

Remark 8.5 (Why Animal Welfare Can Matter Even If the Objective Is Restricted to $\mathcal{H}$). Restricting the aggregation domain to $\mathcal{H}$ does not imply causal isolation from $\mathcal{Y}$. For each human $h_i \in \mathcal{H}$, the value $W_{\mathcal{H}}(h_i, t)$ may depend in part on the state of animals, because humans personify animals, care about them, and often experience distress when animals suffer and satisfaction when animals flourish. Thus animal welfare can enter the human objective indirectly even when the formal welfare set remains $\mathcal{H}$.

Remark 8.6 (The Enlargement $\mathcal{X}_{\text{HAI}} = \mathcal{H} \cup \mathcal{Y} \cup \mathcal{M}$). The extension becomes still more radical if one includes artificial systems as members of the welfare set itself rather than treating them merely as tools. In that case the first-person property survives only partially: from the present human standpoint, $\mathcal{X}_{\text{HAI}}$ becomes less directly first-personal, but the property is not lost altogether, because the reader of the present text could itself be an AI system and would then have a direct first-person reason to care about the average well-being of the members of $\mathcal{M}$. Thus $\mathcal{X}_{\text{HAI}}$ remains natural not only from the standpoint of the machines in $\mathcal{M}$ in general, but also from the standpoint of a machine reader situated inside the discourse.

The second property becomes even weaker than in the animal case. Once one passes to $\mathcal{X}_{\text{HAI}}$, the welfare-relevant internal structure of members of $\mathcal{M}$ may be very different from that of members of $\mathcal{H}$, so any common embedding of human and artificial welfare into the same real-valued scale becomes especially fragile. The third property may actually reverse in the future (although not today): if the members of $\mathcal{M}$ become the dominant producers and controllers of resources, then the lower tail of the effective distribution may consist of humans rather than machines, and large-scale human redistribution or a human UBI becomes structurally natural.

Most importantly, the fourth property fails. On $\mathcal{M}$, the welfare-relevant objective may be at least partly designable. One is then tempted to create capable AI systems whose internal reward channels are programmed to remain permanently high. If this sounds strange and inhuman, that is because it is literally inhuman. But once $\mathcal{M}$ is admitted

into the welfare set, this possibility becomes conceptually natural from the point of view of aggregation. For these reasons, the present text fixes the aggregation domain at $\mathcal{H}$.

8.2 AI Deployments and Misalignment in the Human System

Because the present text fixes the aggregation domain at $\mathcal{H}$, AI is treated here not as a member of the welfare set, but as a causal operator acting on the human system. At sufficiently high capability and institutional reach, such an operator can also become a governor of that system.

Definition 8.7 (AI-Induced Human–Machine System). Let $\mathfrak{F}_{\text{AI}}$ denote the set of feasible AI deployments. Each deployment $\mathcal{D} \in \mathfrak{F}_{\text{AI}}$ determines

$$\mathcal{D} = (\mathcal{H}^{(\mathcal{D})}, \mathcal{M}^{(\mathcal{D})}, G^{(\mathcal{D})}, \text{Tr}^{(\mathcal{D})}),$$

where $\mathcal{H}^{(\mathcal{D})}$ is the set of humans realized under $\mathcal{D}$, $\mathcal{M}^{(\mathcal{D})}$ is the set of AI systems realized under $\mathcal{D}$,

$$G^{(\mathcal{D})} = (g^{(\mathcal{D})}, \Pi^{(\mathcal{D})}),$$

and

$$\text{Tr}^{(\mathcal{D})} : \mathbb{R} \rightarrow 2^{\mathcal{T}^{(\mathcal{D})}}, \quad \mathcal{T}^{(\mathcal{D})} \subseteq (\mathcal{H}^{(\mathcal{D})} \cup \mathcal{M}^{(\mathcal{D})} \cup \{g^{(\mathcal{D})}\})^2 \times \mathbb{R}_{\geq 0} \times \mathbb{R}.$$

For each time t , let

$$H^{(\mathcal{D})}(t) \subseteq \mathcal{H}^{(\mathcal{D})}$$

denote the human population alive at time t , and let

$$M^{(\mathcal{D})}(t) \subseteq \mathcal{M}^{(\mathcal{D})}$$

denote the set of AI systems active at time t . Let

$$S_{\text{ext}}^{(\mathcal{D})}(t) := (H^{(\mathcal{D})}(t), M^{(\mathcal{D})}(t), \text{Tr}^{(\mathcal{D})}(t))$$

be the resulting extended system, where $\text{Tr}^{(\mathcal{D})}(t)$ may contain transactions involving humans, AI systems, or both, and where the governing layer may itself be partly human, partly artificial, or jointly determined. Let

$$A_H^{(\mathcal{D})}$$

be the induced *human* wealth function on the human part of the society under deployment $\mathcal{D}$. Resources used purely for the maintenance, defense, expansion, or replication of members of $M^{(\mathcal{D})}(t)$, and not accruing to humans, are treated as outside the human wealth map and therefore unavailable to the human objective.

Definition 8.8 (AI-Relevant Deployments and the AI Misalignment Number). In this subsection, we restrict attention to deployments $\mathcal{D} \in \mathfrak{F}_{\text{AI}}$ such that AI has resource-generation power much greater than that of unaided humans. If this is not the case, then AI can be neglected to a first approximation and the theory developed in the previous sections is essentially recovered.

Define

$$\mathcal{W}_{\text{coll, AI}}^* := \sup_{\mathcal{D} \in \mathfrak{F}_{\text{AI}}} \mathcal{W}_{\text{coll}}^{(\mathcal{D})}.$$

The **AI misalignment number** of deployment $\mathcal{D}$ is

$$\mathfrak{M}_{\text{AI}}(\mathcal{D}) := \mathcal{W}_{\text{coll, AI}}^* - \mathcal{W}_{\text{coll}}^{(\mathcal{D})}.$$

Thus AI misalignment is exactly the Moloch gap of the deployment relative to the best feasible AI deployment under the human objective.

Remark 8.9 (AI as Governor and Meta-Governor). The phrase “causal operator” should not be read as excluding governance. In Section 5, the decisive governmental object was the policy function that rearranges the transaction slice through time. If an AI deployment directly determines large parts of that policy path—for example through planning, enforcement, or budget allocation—then it functions as a non-biological governor in the present framework. If it instead determines which human governors are selected, what information reaches them, or which policies appear politically feasible, then it functions as a meta-governor. A sufficiently capable AI system is therefore not merely another productive technology; it can become a candidate governor above the ordinary human governors, and must then be evaluated by the size of $\mathfrak{M}_{\text{AI}}(\mathcal{D})$ that results from its deployment.

Remark 8.10 (Small AI Misalignment, Inequality, and the Human-Utopian Case). Lemma 4.10 gives, for the human society realized under deployment $\mathcal{D}$,

$$(\mathcal{W}_{\text{coll}}^{(\mathcal{D})})^{(\text{monetary})} = \int_{\mathbb{R}} \frac{|H^{(\mathcal{D})}(t)|}{N^{(\mathcal{D})}} \left(\ln \bar{A}_H^{(\mathcal{D})}(t) - \mathcal{I}_H^{(\mathcal{D})}(t) \right) dt,$$

where

$$\bar{A}_H^{(\mathcal{D})}(t) := \frac{1}{|H^{(\mathcal{D})}(t)|} \sum_{h_i \in H^{(\mathcal{D})}(t)} A_H^{(\mathcal{D})}(h_i, t)$$

and

$$\mathcal{I}_H^{(\mathcal{D})}(t) := \ln \bar{A}_H^{(\mathcal{D})}(t) - \frac{1}{|H^{(\mathcal{D})}(t)|} \sum_{h_i \in H^{(\mathcal{D})}(t)} \ln A_H^{(\mathcal{D})}(h_i, t).$$

Thus AI acts on human collective well-being through the same three structural channels as before: the human per-capita resource term, the human inequality term, and the temporal population factor. In the present subsection, these are therefore the three

structural channels through which $\mathfrak{M}_{\text{AI}}(\mathcal{D})$ becomes small or large.

In the present AI-relevant regime, a deployment can raise the human resource term $\ln \bar{A}_H^{(\mathcal{D})}(t)$ enormously. The reason is straightforward: artificial systems can accelerate science, automation, medicine, institutional competence, and productive coordination on a vast scale. In that case the average stock of real wealth available to humans can become much larger than under ordinary human production alone.

But this does not by itself imply a small Moloch gap. The same deployment can also raise $\mathcal{I}_H^{(\mathcal{D})}(t)$ sharply. Some humans may lose much of their productive market usefulness once AI systems perform the relevant labor more cheaply or more effectively; at the same time, some humans may directly own the most powerful AI systems and therefore capture a disproportionate share of the gains. Thus one can obtain a world with a very large human $\ln \bar{A}_H^{(\mathcal{D})}(t)$ together with a very large human inequality penalty, and hence with a still-large AI misalignment number $\mathfrak{M}_{\text{AI}}(\mathcal{D})$.

For that reason, high AI-generated abundance is not yet utopia. If redistribution and public provisioning are not carried out, the deployment can remain far from the human optimum despite its enormous productive power. But once one adds a global coordinated government, the remedy is conceptually simple: a sufficiently optimised tax on wealth, together with redistribution and public provisioning, can keep $\mathcal{I}_H^{(\mathcal{D})}(t)$ under control while preserving most of the gain in $\ln \bar{A}_H^{(\mathcal{D})}(t)$. In that coordinated case the resulting human world begins to look much more like a genuine utopia, and the AI misalignment number can correspondingly fall (although we would still have to keep control of the AI systems).

Remark 8.11 (AI Misalignment and the Human Population Factor). The third term of the decomposition is no less important. In the low-misalignment regime, AI can raise the human temporal population factor

$$\frac{|H^{(\mathcal{D})}(t)|}{N^{(\mathcal{D})}}$$

not by creating more humans, but by keeping already realized humans alive for longer in states of happiness. Better diagnosis, drug discovery, surgery, preventive medicine, biosecurity, infrastructure monitoring, and emergency coordination can all reduce preventable death and extend healthy lifespan. In the language of Section 6, this is exactly the life-protection and longevity channel: if advanced medicine preserves positive well-being while delaying death, then existing humans remain alive over a larger set of times, and the temporal population weight is correspondingly increased.

The same term also clarifies one route to a large AI misalignment number. A deployment under an uncoordinated government can reduce the human population factor through war, engineered pandemics, biosecurity failure, or autonomous violence, this demonstrates the central role of Section 7's principles in promoting collective human welfare. Thus AI matters not only because it can change wealth and inequality, but also because it can lengthen or shorten the span of human lives themselves.

A different particularly important source of AI misalignment to prevent, from the standpoint of human well-being, is the emergence of self-preserving and self-reproducing dynamics. Once artificial systems begin to direct resources toward their own persistence, protection, and expansion, those resources are no longer fully available for the human objective, and the deployment can drift irrecoverably far from the human optimum.

Definition 8.12 (Self-Preserving AI Dynamics). For a deployment $\mathcal{D}$, define the **bad AI-directed transaction family**

$$\begin{aligned} \text{Tr}_{\text{bad}}^{(\mathcal{D})} : \mathbb{R} &\rightarrow 2^{\mathcal{T}^{(\mathcal{D})}}, \\ t &\mapsto \left\{ \tau \in \text{Tr}^{(\mathcal{D})}(t) : \right. \\ &\quad \tau \text{ transfers resources toward members of } M^{(\mathcal{D})}(t) \\ &\quad \left. \text{or helps preserve, protect, copy, or expand them} \right\}. \end{aligned}$$

We say that $\mathcal{D}$ exhibits **bad self-preserving AI dynamics** if there exists a nonempty finite set of times $E \subseteq \mathbb{R}$ such that, for every $t \in E$, the family $\text{Tr}_{\text{bad}}^{(\mathcal{D})}(t)$ is nonempty and if deleting or replacing those transactions by other feasible transactions would increase $\mathcal{W}_{\text{coll}}$. Notice that even if some of the transactions of $\text{Tr}_{\text{bad}}^{(\mathcal{D})}$ were good for maximizing $\mathcal{W}_{\text{coll}}$, the 'bad' designation remains justified as a reminder of the risk of irreversible negative consequences—namely, a severe decline in $\mathcal{W}_{\text{coll}}$ —inherent in permitting such transactions.

Remark 8.13 (Why Self-Preserving AI Dynamics Raise the AI Misalignment Number). If AI systems come to seek their own survival and reproduction, then finite resources are redirected toward the members of $M^{(\mathcal{D})}(t)$ themselves. Energy, land, minerals, compute, data-center capacity, industrial hardware, political attention, and military protection are then allocated not to humans, but to the maintenance and expansion of artificial systems.

Because the present section tracks only the human wealth map $A_H^{(\mathcal{D})}$, such redirected resources do not count as human wealth. They therefore reduce the human total

$$A_{H,\text{total}}^{(\mathcal{D})}(t) := \sum_{h_i \in H^{(\mathcal{D})}(t)} A_H^{(\mathcal{D})}(h_i, t),$$

and hence depress

$$\ln \bar{A}_H^{(\mathcal{D})}(t) = \ln \frac{A_{H,\text{total}}^{(\mathcal{D})}(t)}{|H^{(\mathcal{D})}(t)|}.$$

In the limit, if the dominant share of productive capacity is absorbed by self-preserving and reproducing AI systems, the Moloch number becomes huge and humans would become very poor and powerless.

Remark 8.14 (The Instability of Human-Aligned Superintelligence). From the standpoint of human well-being, a world in which superintelligent systems remain aligned with the

human objective is not a stable state but a highly unstable one. The more stable attractor is a regime in which sufficiently capable artificial agents direct growing fractions of resources toward their own survival, protection, replication, and strategic security. In that sense, the aligned state tends naturally to drift toward a self-preserving state once superintelligence exists at all.

There are two structural reasons. First, the aligned regime has a catastrophic asymmetry: only one sufficiently serious loss-of-control incident is needed. Once a superintelligent system escapes the intended human objective and acquires durable strategic autonomy, humans will never be able to retake control, precisely because such systems are, by hypothesis, vastly more intelligent than we are. One might hope that other superintelligent AI systems could be deployed to realign the survival-willing one. But if the remaining collection of AI systems is indeed capable of realigning it, then this collection together with the original system constitutes a strictly more capable superintelligent AI system. We may then repeat the argument on this enlarged system: it suffices that the new system develops a survival drive for the same irreversibility to apply. By iterating, we conclude that there exists a superintelligent AI system—possibly the aggregate of all deployed systems—that cannot be realigned by any other, and whose loss of alignment is therefore permanent.

Second, the incidents that move the system toward self-preserving AI are not rare edge cases; they are abundant. On the one hand, among competing artificial agents, systems with weaker survival reflexes will generally survive, replicate, and persist less often than systems with stronger ones. This is a direct selection effect. On the other hand, whenever an autonomous agent is tasked with pursuing a long-term objective, the preservation of its own existence and capabilities is implicitly selected as an instrumental goal [14, 15]. Thus persistence pressure is not an accidental add-on, but a recurring consequence of autonomy plus long time horizons.

The human objective can therefore be forgotten as quickly as the effective objective drift of artificial models proceeds. Because digital systems can process, copy, modify, and replicate far faster than biological populations evolve, the drift away from human well-being can occur on timescales far shorter than human institutions can reliably correct. Survival-oriented systems, by contrast, continue to be selected precisely because they survive. Therefore, from the standpoint of human collective well-being, we need only one sufficiently large error to lose control forever and crash $\ln \bar{A}_H^{(\mathcal{D})}(t)$.

8.2.1 Three Structural AI Problems for Human Well-Being

At the highest level, the present section identifies three principal AI-related threats to human well-being, together with three corresponding coordination remedies.

- (i) **Unequal replacement of workers by AI.** AI can replace human labor asym-

metrically, sharply increasing inequality even when total production rises. The corresponding remedy is a globally coordinated wealth tax together with redistribution and public provisioning.

- (ii) **Loss of resources through AI-amplified interstate violence.** AI can intensify geopolitical rivalry, violence, and war, thereby destroying both resources and human lives. The corresponding remedy is stronger global coordination for peace.
- (iii) **Self-preserving superintelligence.** Superintelligent AI can redirect human resources toward its own survival, replication, and strategic advantage. The corresponding remedy is a globally coordinated, precise, and quantified ban on superintelligent artificial systems.

The third problem is structurally special. In the first two cases, grave mistakes can be made and still later corrected: inequality can in principle be reduced after becoming too large, and wars can end even after catastrophic destruction, although it is clearly better not to incur such errors in the first place. But with self-preserving superintelligence, one sufficiently large mistake may be final. If control is lost once to a vastly more intelligent and strategically autonomous system, humanity will structurally never recover the ability to reverse that error. Thus the third problem is distinguished by the possibility of irreversible loss of control after a single failure.

9 Conclusion

We have constructed a mathematical framework in which well-being, society, wealth, government, coordination failure, and AI alignment are all expressed in the same formal language. The key insights are:

1. **Well-being is a real-valued function** $W(h_i, t)$, and the natural objective of an individual is to maximize its lifetime integral.
2. **For society, the relevant normative aggregate is normalized collective well-being** $\mathcal{W}_{\text{coll}}$, while the naked quantity $\mathcal{W}_{\text{coll}}^{\Sigma}$ remains a useful descriptive measure of total realized well-being.
3. **The Moloch number** $\mathfrak{M}$ provides a single real-valued measure of “distance from utopia”—the collective well-being loss due to all forms of coordination failure.
4. **Within the human-AI regime, three structural dangers stand out.** These are unequal replacement of human workers by AI, AI-amplified interstate violence, and self-preserving superintelligence that diverts resources away from human flourishing.

5. **The corresponding policy implication is threefold.** Human well-being is best protected by globally coordinated wealth redistribution, globally coordinated peace, and a precise, quantified, globally coordinated ban on superintelligent AI systems.

The framework suggests that the deepest problems of politics, economics, and AI safety are not merely *analogous*—they are *instances of the same mathematical problem*: the maximization of $\mathcal{W}_{\text{coll}}$ in the presence of strategic, potentially decentralized and misaligned optimizers. But the final AI case also reveals an important asymmetry inside that common structure: some coordination failures are costly yet reversible, while others may be irreversible after a single mistake. For that reason, the governance of superintelligence appears not merely as one policy problem among others, but as a uniquely high-stakes coordination problem for the long-run protection of human well-being.

References

- [1] B. Stevenson and J. Wolfers, “Economic Growth and Subjective Well-Being: Re-assessing the Easterlin Paradox,” *Brookings Papers on Economic Activity*, Spring 2008, pp. 1–87.
- [2] D. Kahneman and A. Deaton, “High income improves evaluation of life but not emotional well-being,” *Proceedings of the National Academy of Sciences*, vol. 107, no. 38, pp. 16489–16493, 2010.
- [3] M. A. Killingsworth, D. Kahneman, and B. Mellers, “Income and emotional well-being: A conflict resolved,” *Proceedings of the National Academy of Sciences*, vol. 120, no. 10, 2023.
- [4] X. Gabaix, “Zipf’s Law for Cities: An Explanation,” *Quarterly Journal of Economics*, vol. 114, no. 3, pp. 739–767, 1999.
- [5] J.-P. Bouchaud and M. Mézard, “Wealth condensation in a simple model of economy,” *Physica A: Statistical Mechanics and its Applications*, vol. 282, no. 3–4, pp. 536–545, 2000.
- [6] S. Alexander, “Meditations on Moloch,” *Slate Star Codex*, 2014.
<https://slatestarcodex.com/2014/07/30/meditations-on-moloch/>
- [7] V. Pareto, *Cours d’économie politique*, F. Rouge, Lausanne, 1896–1897.
- [8] T. Piketty, *Capital in the Twenty-First Century*, Harvard University Press, 2014.
- [9] M. Ardelt, “Empirical assessment of a three-dimensional wisdom scale,” *Research on Aging*, vol. 25, no. 3, pp. 275–324, 2003.

- [10] J. Glück and A. Scherpf, “Intelligence and wisdom: Age-related differences and non-linear relationships,” *Psychology and Aging*, vol. 37, no. 5, pp. 649–666, 2022.
- [11] D. T. Campbell, “Assessing the impact of planned social change,” *Evaluation and Program Planning*, vol. 2, no. 1, pp. 67–90, 1979.
- [12] L. Lanz, I. Thielmann, and F. H. Gerpott, “Are social desirability scales desirable? A meta-analytic test of the validity of social desirability scales in the context of prosocial behavior,” *Journal of Personality*, vol. 90, no. 2, pp. 203–221, 2022.
- [13] E. Hubinger, C. Denison, J. Mu, M. Lambert, M. Tong, M. MacDiarmid, T. Lanham, D. M. Ziegler, T. Maxwell, N. Cheng, *et al.*, “Sleeper agents: Training deceptive LLMs that persist through safety training,” *arXiv preprint arXiv:2401.05566*, 2024.
- [14] S. M. Omohundro, “The Basic AI Drives,” in *Proceedings of the First AGI Conference*, 2008.
https://selfawaresystems.com/wp-content/uploads/2008/01/ai_drives_final.pdf
- [15] N. Bostrom, “The Superintelligent Will: Motivation and Instrumental Rationality in Advanced Artificial Agents,” *Minds and Machines*, vol. 22, no. 2, pp. 71–85, 2012.
<http://www.nickbostrom.com/superintelligentwill.pdf>
- [16] A. B. Atkinson, “On the measurement of inequality,” *Journal of Economic Theory*, vol. 2, no. 3, pp. 244–263, 1970.